MH-I-1751/5
August 7, 1973

FEDERAL SPECIFICATION

INSULATION SLEEVING, THERMAL, PIPE COVERING (CALCIUM SILICATE, FOR TEMPERATURES UP TO 1500°F.)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers asbestos free calcium silicate pipe covering thermal insulation intended for use on surfaces which reach temperatures up to 1500°F. The actual temperature limit shall be as specified (see 6.2).

1.2 Classification.

1.2.1 Type and classes. Insulation shall be furnished in the following type and classes, as specified (see 6.2):

Type I - Pipe insulation.
Class I - Uncovered.
II - Aluminum jacketed.
III - Kraft paper, aluminum foil jacketed.
IV - Dual temperature jacketed.
V - Canvas jacketed.
VI - Other jackets.

1.3 The complete requirements for procuring thermal insulation sleeving described herein shall consist of this document and the latest issue of specification MH-I-1751/GEN and documents referenced therein.

2. APPLICABLE DOCUMENTS

2.1 The latest issue of Federal Specification MH-I-1751/GEN and documents referenced therein.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

C 356 - Test for Linear Shrinkage of Preformed High Temperature Thermal Insulation Subjected to Soaking Heat.
C 533 - Specification for calcium Silicate Block and Pipe, Thermal Insulation.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUUIREMENTS

3.1 Material. Calcium silicate asbestos free thermal insulation shall be composed principally of hydrous calcium silicate with reinforcing fiber, and unless otherwise specified shall meet the requirements of ASTM C 533, except for Quality Assurance, Packing, Marking and Packaging requirements.

3.1.1 Construction. Calcium silicate pipe insulation shall be supplied either as hollow cylindrical shapes split in half length wise (in a plane including the cylindrical axis) or as curved segments. The pipe insulation shall be furnished in sections or segments in a length of 36 inches, to fit sizes of pipe in accordance with table I, and in nominal thicknesses of from 1 to 3 inches, in 1/2 inch increments.

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TABLE I. Thickness of hollow cylinder pipe insulation

Pipe Size		Nominal thickness											
		1 inch		1-1/2 inch		2 inches		2-1/2 inches		3 inches		4 inches	
Nominal diameter	Outer diameter	Actual thickness		Actual thickness		Actual thickness		Actual thickness		Actual thickness		Actual thickness	
Inches	Inches	Inches		Inches		Inches		Inches		Inches		Inches	
1/2	0.840	1		1-9/16		2-1/16		2-7/8		3-3/8		4-3/8	
3/4	1.050	1		1-7/16		1-15/16		2-3/4		3-1/4		4-1/4	
1	1.315	7/8		1-9/16		2-3/32		2-5/8		3-1/8		4-1/8	
1-1/4	1.660	1-1/16		1-5/8		1-29/32		2-7/16		2-15/16		3-15/16	
1-1/2	1.900	7/8		1-1/2		2-5/16		2-13/16		3-5/16		4-3/8	
2	2.375	1		1-9/16		2-3/32		2-19/32		3-3/32		4-5/32	
2-1/2	2.875	1-1/32		1-27/32		2-11/32		2-27/32		3-11/32		4-13/32	
3	3.500	1		1-17/32		2-1/32		2-17/32		3-1/32		4-3/32	
3-1/2	4.000	1-9/32		1-25/32		2-9/32		2-25/32		3-11/32		4-11/32	
4	4.500	1-1/32		1-17/32		2-1/32		2-17/32		3-3/32		4-3/32	
4-1/2	5.000	1-9/32		1-25/32		2-9/32		2-27/32		3-11/32		4-15/32	
5	5.563	1		1-1/2		2		2-9/16		3-1/16		4-3/16	
6	6.625	31/32		1-15/32		2-1/32		2-17/32		3-1/32		4-5/32	
7	7.625	-----		1-17/32		2-1/32		2-17/32		3-5/32		4-5/32	
8	8.625	-----		1-17/32		2-5/32		2-21/32		3-5/32		4-5/32	
9	9.625	-----		1-17/32		2-3/32		2-19/32		3-3/32		4-3/32	
10	10.750	-----		1-19/32		2-3/32		2-19/32		3-3/32		4-3/32	
11	11.750	-----		1-19/32		2-3/32		2-19/32		3-5/32		4-5/32	
12	12.750	-----		1-9/16		2-1/16		2-9/16		3-1/16		4-1/16	
14	14.000	-----		1-7/16		1-15/16		2-7/16		2-15/16		3-7/16	
Over 14 up to and including 33	-----	-----		1-7/16		1-15/16		2-7/16		2-15/16		3-7/16	

Pipe Size		Nominal thickness							
		4-1/2 inches		5 inches		5-1/2 inches		6 inches	
Nominal diameter	Outer diameter	Actual thickness		Actual thickness		Actual thickness		Actual thickness	
Inches	Inches	Inches		Inches		Inches		Inches	
1/2	0.840	4-15/16		5-5/16		5-7/16		5-15/16	
3/4	1.050	4-13/16		5-3/16		5-5/16		5-15/16	
1	1.315	4-11/16		5-1/16		5-3/16		5-13/16	
1-1/4	1.660	4-17/32		5-1/32		5-1/32		5-13/32	
1-1/2	1.900	4-29/32		5-13/32		5-5/32		5-15/32	
2	2.375	4-21/32		5-5/32		5-17/32		6-1/32	
2-1/2	2.875	4-29/32		5-17/32		5-7/32		6-1/32	
3	3.500	4-19/32		5-7/32		5-9/32		6-3/32	
3-1/2	4.000	4-11/32		4-31/32		5-11/32		6-3/32	
4	4.500	4-23/32		5-7/32		5-11/32		6-5/32	
4-1/2	5.000	4-15/32		4-31/32		5-13/32		6-5/32	
5	5.563	4-11/16		5-3/16		5-15/16		6-7/32	
6	6.625	4-21/32		5-7/32		6-1/32		6-9/32	
7	7.625	4-19/32		5-9/32		6-3/32		6-11/32	
8	8.625	4-19/32		5-11/32		6-5/32		6-13/32	
9	9.625	4-21/32		5-13/32		6-7/32		6-15/32	
10	10.750	4-19/32		5-15/32		6-9/32		7-1/32	
11	11.750	4-21/32		5-17/32		6-11/32		7-3/32	
12	12.750	4-21/32		5-19/32		6-13/32		7-5/32	
14	14.000	4-19/32		5-21/32		6-15/32		7-7/32	
Over 14 up to and including 33	-----	4-19/32		5-23/32		6-17/32		7-9/32	

Note. The inner diameter of pipe insulation listed in table I approximates the outer diameter of iron pipe sizes. The insulation can be applied either to a pipe or as a second layer to a smaller size of pipe insulation.

1/ Pipe insulation may be furnished in two layers to form total required thickness where the nominal thickness required is in excess of 2 inches.

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3.1.2 Classes.

3.1.3 Class I. Uncovered pipe thermal insulation shall be furnished without jackets (see 6.2).

3.1.4 Class II. Aluminum jackets shall be fabricated of 16 mil (0.016 inch) thick, 3003 alloy, having either a smooth or embossed surface. The aluminum shall be formed to fit the cylinder and have a lock seam which permits opening closing, and locking about the cylinder. The cylinder shall have a moisture-barrier securely attached with sufficient overlap to prevent moisture migration between the insulation and the aluminum jacket. An aluminum butt strap shall be furnished with each aluminum jacket. Butt straps shall be formed in a manner which will resist penetration and degradation of the insulation by windborne water.

3.1.5 Class III. A laminated jacket with kraft paper outer surface bonded to a vapor barrier film shall be furnished. The jacket shall have permeability rating of .02 perms, and a tensile strength of not less than 30 lbs. per inch - width, and shall have a beach puncture resistance of not less than 50 beach units. No jacket shall be furnished on the inner layer of double layer construction.

3.1.6 Class IV. Dual temperature jackets shall be three ply laminates of kraft paper, foil, and kraft paper. The weight of paper, thickness of aluminum foil, and water vapor transmission rate and beach puncture resistance shall be limited as stated for class III.

3.1.7 Class V. Canvas jacketed pipe insulation shall have a cotton cloth jacket securely attached to the cylinder in a manner which provides a minimum overlap of 1 inch on one end and along the cylinder seam.

3.1.8 Class VI. Other jackets shall be fabricated of asphalt saturated asbestos felts, asphalt saturated and coated asbestos felts, polyvinyl fluoride (PVF) laminated neoprene, neoprene saturated asbestos felt, and other plastic film laminates.

3.1.9 Bands. Aluminum or steel bands, black japanned or lacquered, shall be furnished when specified and as specified (see 6.2). All bands shall be at least 1/2-inch wide and not less than 0.005-inch thick. Bands shall be furnished at the rate of five per two sections of pipe covering.

3.2 Physical requirements.

3.2.1 Thermal conductivity. When tested in accordance with 4.4.1 of MH-I-1751/GEN the thermal insulation sleeving K factors shall be in accordance with table II.

TABLE II. Thermal conductivity

Thermal conductivity Max., Btu-In/Hr. Sq. Ft. F.	
At Mean Temperature	
300	0.50
500	0.60
700	0.71
Density, Max., Pcf	14.0

3.2.2 Fire hazard classification. When tested in accordance with 4.4.2 of MH-I-1751/GEN the jackets for thermal insulation sleeving shall have a flame spread rating not greater than 25. The smoke developed rating shall not be greater than 50.

3.2.3 Linear shrinkage. When tested in accordance with 4.4.1 of this specification at the temperature rating specified in the contract or purchase order, (see 6.2 and 6.4) thermal insulation sleeving shall show no physical changes that adversely affect its service qualities.

3.2.4 Temperature ratings. When tested in accordance with 4.4.6 of MH-I-1751/GEN rating specified in the contract or purchase order (see 6.2 and 6.4) thermal insulation sleeving shall show no physical changes that adversely affect its service qualities.

3.2.5 Moisture adsorption. When tested in accordance with 4.4.7 of MH-I-1751/GEN the thermal insulation sleeving for use below ambient shall gain no more than 2.5 percent by volume.

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4. QUALITY ASSURANCE PROVISIONS

4.1 The applicable Quality Assurance Provisions of Federal Specification HH-I-1751/GEN and the documents therein, shall apply with the following additional requirements.

4.2 Sampling.

4.2.1 In accordance with 4.2.1 of HH-I-1751/GEN, unless otherwise speicified (see 6.2).

4.3 Examination.

4.3.1 End item. Insulation selected in accordance with 4.2.2 of HH-I-1751/GEN shall be examined for defects listed in table I of HH-I-1751/GEN.

4.4 Test methods. Determination of the fire rating shall be in accordance with ASTM E 84 (see 3.2.2).

4.4.1 Linear shrinkage. Determination of linear shrinkage shall be in accordance with ASTM C 356 (see 3.2.3).

5. PREPARATION FOR DELIVERY. In accordance with Federal Specification HH-I-1751/GEN and documents referenced therein.

6. NOTES

6.1 Intended use. Thermal pipe insulation covered by this specification are intended for insulating surfaces up to 1500°F.

6.2 Ordering data. Purchaser should select the preferred options permitted herein and included the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class as applicable for pipe insulation (see 1.2.1).
(c) Other jacket material required when class 5 is specified (see 3.1.7).
(d) Size and thickness of pipe insulation.
(e) Size of lot, if different from 4.2.1, of HH-I-1751/GEN.
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2) of HH-I-1751/GEN.
(g) Type bands required (see 3.9).

MILITARY CUSTODIANS:

Army - ME
Navy - YD
Air Force - 84

Review activity:

Navy - SH

User activity:

Coast Guard - CG

Military Coordinating Activity:

Army - ME

Preparing activity:

GSA-FSS

Civil Agency Coordinating Activities:

COMMERCE - NBS
DOT - FIS
GSA - FSS, PCD, TCS
INTERIOR - BPA

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